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Novel Foam-Assisted Gas Lift Technology for Resuscitation of High-Salinity and High-GOR Oil Wells

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Abstract

Objectives/Scope: Liquid loading is a common problem in mature gas fields; furthermore, it has ability to cease the deliverability of oil wells and has significantly affected ultimate recovery from existing oil fields. This paper outlines a strategy for the successful implementation of a distinctive approach: foam-assisted gas lift technology to resuscitate high-salinity and high-GOR oil wells with noteworthy production uplift in one of the largest offshore oil fields located in middle eastern basin.

Methods, Procedures, Process: This rig-less technique incorporates introduction of solid and liquid foaming agents in conjunction with gas lift which not only overcome the limitation of gas lift systems but also transformed well performance. Formulation of well selection criteria, individual well sampling and assessment, proper data gathering, data analysis, application methodology and custom-made design of de-liquification program have made it possible to re-activate decades-old wells to enhance production from depleted reservoirs with liquid loading problems.

Results, Observations, Conclusions: A pilot project on five wells of a major operator in the middle east exhibited a significant improvement in well performance with an incremental gain of 3000+ bopd along with savings of 10 mmscfd of injection gas from an oil reservoir which had ceased to flow due to high water cut or unavailability of enough reservoir support. Application of this technology has broken the barrier in evaluation of these wells and provides an effective means of reviving the oil wells with liquid loading issues beyond the capability of standard de-liquification methods. It is a cost-effective technology to unravel the untapped reserves from existing wells.

Novel/Additive Information: This paper considers multiple scenarios that help revive and elevate a potential hydrocarbon source and recommends suitable actions which allow production engineers to analyze performance of reviving oil wells. The paper will also discuss the role of this technology in the overall production enhancement workflow.

With the increased complexity of mature fields, each next step to improve recovery comes with increased cost. Therefore, novel horizons of affordable technology will play a pivotal role in maximizing economic recovery and meet rising demand in the current business environment.

Introduction

When gas flows to surface, the gas will carry the produced liquids to the surface if the gas velocity is high enough. A high gas velocity generates a mist flow pattern in which the liquids are finely dispersed in the gas.

This results in a low liquid holdup and, as a result, a low pressure drop due to the hydrostatic component of the flowing fluids. As the velocity of the gas in the production conduit drops with time, the velocity of the liquids carried by the gas decreases even faster. As a result, liquids begin to collect on the walls of the conduit, liquid slugs begin to form, and eventually liquids accumulate in the bottom of the well, adding to the percent of liquids in the conduit while the well is flowing. The presence of more liquids in the production conduit while the well is flowing can slow or even stop gas production altogether.

Liquid loading of a gas well is the inability of the produced gas to remove the produced liquids from the wellbore. Under this condition, produced liquids will accumulate in the wellbore, leading to reduced production and shortening of the productive lifespan until the well no longer will produce.

At the core of liquid loading is a balance—or imbalance—between gas flow energy and the gravitational hold-up of liquids in the wellbore.

Turner's Critical Velocity Concept

Turner et al. (1969) established a practical framework for predicting the onset of liquid loading using a critical gas velocity model:

$$v_c = C \sqrt{\frac{\sigma(\rho_l - \rho_g)}{\rho_g^2}}$$

Where:

- v_c = critical gas velocity required to lift liquids (ft/s or m/s)
- C = flow-dependent constant (typically ~ 0.35 for vertical flow)
- σ = liquid-gas surface tension (dyne/cm)
- ρ_l, ρ_g = densities of liquid and gas respectively (lb/ft³ or kg/m³)

When the actual gas velocity falls below v_c , liquids are no longer transported upward efficiently and begin to accumulate. This results in increased hydrostatic pressure, reduced well deliverability, and, eventually, well loading.

Liquid loading can lead to erratic, slugging flow and decreased production. The well may eventually die if the liquids are not continuously removed. Often, as liquids accumulate in a well, the well simply produces at a lower rate than expected. If the gas rate is high enough to remove most or all of the liquids, the flowing tubing pressure at the formation face and production rate will reach a stable equilibrium. The well will produce at a rate that can be predicted by the reservoir inflow production relationship (IPR) curve.

It is a prevalent problem which comes in the life cycle of a gas well. If the gas rate is too low, the pressure gradient in the tubing becomes large due to the liquid accumulation, resulting in increased pressure on the formation. As the backpressure on the formation increases, the rate of gas production from the reservoir decreases and may drop below the critical rate required to remove the liquid. More liquids will accumulate in the wellbore and the increased bottomhole pressure (BHP) will further reduce gas production and may even kill the well.

The key root causes of Liquid Loading are as follows: the susceptibility of well to liquid loading is influenced by a combination of reservoir, fluid, and wellbore parameters:

Reservoir Depletion

As reservoir pressure declines, the driving force for gas production weakens. Lower gas rates often lead to insufficient velocity to lift liquids, particularly in tight or unconventional plays.

Increased Liquid Production

Produced water (formation or completion fluids) and condensates contribute directly to the liquid column that must be lifted. Higher water-gas ratios (WGRs) accelerate the onset of loading.

Large Tubing Diameters

Wells completed with large-diameter tubing ($>2\text{-}3/8''$) may see gas velocities drop below critical thresholds more quickly, especially as rates decline.

High Surface Tension and Fluid Viscosity

High surface tension increases the difficulty of droplet entrainment, and viscous fluids resist movement, compounding the loading effect.

Deviated or Horizontal Wellbores

In deviated or horizontal sections, liquids may accumulate in low spots or undulations, leading to slug flow or bridging—making deliquification even more challenging.

Accurate diagnosis is essential before implementing any mitigation strategy. Effective surveillance includes:

- **Production Trend Analysis:** A flattening or decline in gas rate with rising tubing pressure is a strong loading indicator.
- **Critical Velocity Modeling:** Comparison of actual velocity vs. Turner critical velocity.
- **Pressure Transient Testing:** Can identify elevated skin or transient slugging behavior.
- **Flow Regime Identification:** Use of multiphase simulators to model intermittent vs. annular flow.
- **Production Logging Tools (PLT):** Provide in-situ liquid holdup and phase identification.
- **Acoustic Liquid Level Surveys:** Non-intrusive method to estimate liquid level in casing.

Predominantly an integrated approach is recommended encouraged to apply during surveillance and mitigation framework. This includes:

- Regular monitoring of critical velocity thresholds
- Periodic well testing and production logging
- Integration of nodal analysis with real-time production data
- Predictive modeling using transient multiphase simulators

Economic evaluation is critical to determine the viability of intervention techniques, especially for marginal wells with limited reserves.

Case Study

Background Field History

Two decades ago, in a challenging offshore environment, a significant number of wells were completed with and without gas lift completion. Over the period, many of them quit flowing and were shut in for 10 to 18 years, due to higher water cut or insufficient reservoir support.

A detailed technical feasibility study was performed, which incorporated six different technologies to identify the optimum deliquefying method to resuscitate the oil wells to unlock the untapped potential by minimizing well intervention with the most economically feasible technology. The screening criteria below were used to assess these technologies.

Table 1—Screening criteria of technologies

Parameters	Foam-Assisted Lift	Plunger Lift	Insert String	Capillary Foam	Velocity String	Nitrogen Lift
Rigless Operation	Yes	Yes	Yes	Medium	No	No
Slickline Intervention	Yes	Yes	Yes	Medium	No	No
HSE Threat	No	No	No	Yes	Yes	Yes
Cost Effective	Yes	Yes	Yes	No	No	No
Complexity	No	Medium	Medium	Yes	Yes	Yes

Foam-assisted lift was selected to execute a pilot project on 5 wells as it meets the above norms.

Selection of well candidates

Well candidates for this are selected based on the criteria below:

Table 2—Well candidates selection criteria

Parameters	Foam-Assisted Gas Lift Wells
Gross Liquid (bpd)	>500
BS&W %	<90
GOR (scf/stb)	>2500
Net Oil (BOPD)	>100
Fluid Level (feet)	< 1000 ft below deepest GLV
Down Hole Temperature at FL depth (F)	>100
Down hole pressure at FL depth (PSI)	>500
Fluid Sample (PPM)	<250
Gas Lift Injection (MMSCFD)	1.5

In case of gas lifted wells, primarily foam sticks were planned to kick-off and unload the water, wells will be produced from the existing gas lift completions for a sustainable long-term production.

Selection of Foam Sticks

Based on the water cut, well water sample salinity analysis, and crude oil compositional analysis, the below two types of foam sticks were selected:

- Oil Foam Sticks: These are condensate-dispersible water-soluble sticks that contain surfactants that foam both the condensate and water column in gas-condensate wells.
- Oil and Water Foam Sticks: These are hybrid sticks which contain hydrocarbon and water-soluble components, they are used to remove condensate and water from gas-condensate wells and flowing oil wells.

In addition, liquid foamer was selected to create foaming action by injecting through the surface pump into the tubing string via the existing gas lift completion.

To select an optimum type of foam stick and estimate the quantity, laboratory compatibility tests between collected reservoir fluid samples and selected agents were performed to evaluate the effectiveness, height of foam and stability of foam action in downhole conditions. During laboratory testing, the desired agitation was created by vigorously shaking beaker after the foam stick dissolves in the sample. This indicates the importance of injecting gas to enhance the foaming action at the well site.

Well X

Well History

This well was completed in a deep hydrocarbon reservoir located in a mature offshore oilfield with a low BHP. It had a gross liquid production of 1050 bpd with 25% water cut before it ceased to flow in 2004. Last measured fluid level 2026 ft, 121°F and 705 psi at fluid level depth with 240,000 ppm salinity.

It was completed with 3.5in. production tubing in a 7in. production liner. The reservoir has a good productivity index with flowing bottom hole pressure of 3000 psi. The produced fluid is light 39° API oil without any wax content. The formation gas to oil ratio (GOR) value is 9500. Gas lift was selected as the preferred artificial lift technology due to availability of essential infrastructure facilities and the crude oil physical and chemical properties.

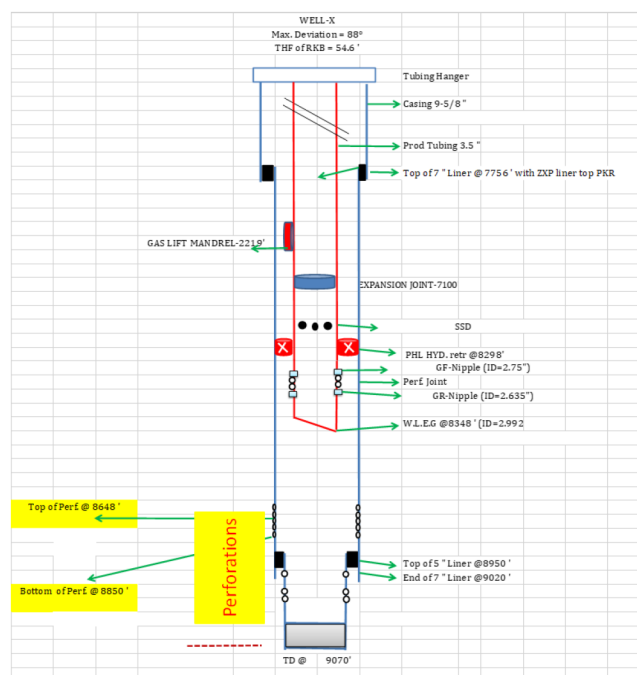


Figure 1—Well completion diagram

Job Planning

Identification of Well Performance. Analysis of well performance has been conducted by using a software package to simulate the well conditions and to depict the behavior of petroleum fluids in this well.

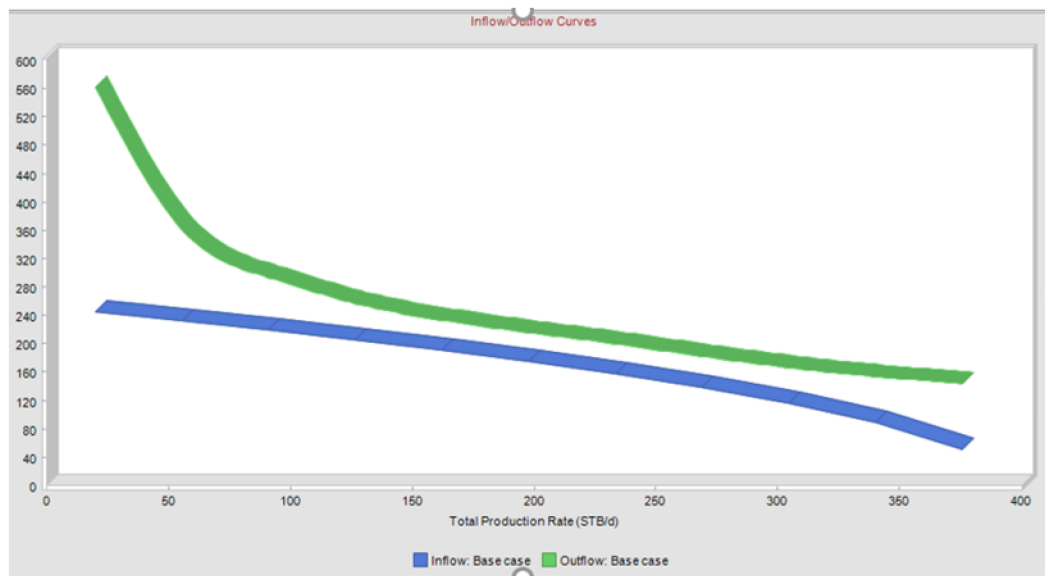


Figure 2—Operating point analysis

As it can be seen from the above plot, inflow and outflow curves do not intersect, so reservoir pressure is insufficient to produce fluids to surface by natural flow.

Foaming agent calculation. The data supplied show this well has approximately 56 barrels of fluid in the vertical section with 25% BS&W. For this, we would recommend using the oil foam sticks.

Based on the calculation, this well would require 20-30 oil foam sticks per application; possible applications would be between 2 and 4. This will depend on the reaction we see at the surface in the well and the wellhead pressure.

In conjunction with this, we need to inject 10-15 liters of liquid foamer surfactant over a two-hour period into the gas lift system.

Development and Execution of Field Operating Procedure. Even though field operating procedures will change from well to well, reservoir to reservoir and from field to field, the below key elements are enlisted to kick-off a high GOR and high salinity well completed with a gas lift.

- Topside Setup: Bleed-off the flow lines and line-up the well to the test separator with zero back pressure. Connect the chemical-injection pump on the lift gas injection line to inject liquid foamer. Pressure gauges will record the real time behavior of tubing head and casing head pressure.
- Downhole Setup: Identify the test rack opening pressure valves of IPO valves.

Evaluate the condition of gas lift valves, changing of old valves are essential to eliminate the possibility of valve blockage or failure.

Recognize the type of downhole safety valve (i.e. tubing retrievable or wireline retrievable), minimum restriction ID etc. Tagging fluid level through a blind box via SL or via a non-invasive way through an echometer shot in tubing string to estimate the reservoir deliverability and verify the number of foam sticks required to unload the well. Drifting of the tubing string is vital to ascertain no obstruction is in the path of the foam sticks.

- Determination of well performance: Based on the last available well test parameters, it is indispensable to estimate the inflow and outflow performance to identify how much drawdown would be required to establish flow in the current setup. It will also assist to discover the other potential alternative deliquification technique to bring life in a shut-in well.

Calculation of number of foam sticks and volume of liquid foaming agent.

These steps are involved during the execution phase at the well site:

1. Shut in the well.
2. Complete standard SL rig up including a single/dual ram SL BOP and lubricator on the wellhead tree adapter.
3. Place 15-20 oil/water foam sticks into the lubricator. If the lubricator is too small to accommodate the sticks, place as many sticks into the lubricator as possible.
4. Just drop the sticks or place them at the mouth of the lubricator and drop them.
5. Once the sticks have been placed in the lubricator, replace the weco cap.
6. Open the master valve, then open the top valve of the wellhead. The sticks should then fall into the production tubing. A tap with a bronze hammer can be used in case the sticks have hung up in the lubricator.
7. Depending on the size of the lubricator, this may have to be performed at least 2 or 3 times. The lubricator must have a pressure gauge, and a bleeder valve installed.
8. Run the appropriate size gauge cutter for drifting the tubing to avoid any obstruction.
9. The well should be left for a minimum of 60-90 minutes to allow the sticks to dissolve.
10. Start the gas lift injection with a pre-determined gas injection rate (It may take 1-1½ hours to open the gas lift valves). The injected gas will also assist in agitating the formed foam and enhance the foaming mechanism, which will eventually decrease the hydrostatic head.
11. Once the GLV's are open, start the liquid foam injection (20–30 liters).
12. Observe WHP pressure. Once it has reached its normal pressure, open the well very slowly for 5-10 minutes with a half turn every 30 seconds, and then open the well to maximum choke. Never let the well bleed down to zero. Close the well when the pressure drops to 30% of the initial WHP.
13. Observe the well offloading. The well may offload a little; try not to let the well die again. If the well is offloading or dies again, or if the rate drops and the pressure drops to about 30% of its normal pressure, shut the well in again.
14. This cycle can be performed 2 to 3 times until the well flows on its own.
15. Once the well starts offloading, keep the well flowing.

After 24 hours of the well flowing without the use of foam sticks, you can try and put it into the main production header.

Results

Well performance has been increased significantly with an increment of more than 1030 bbls of oil per day from a high salinity and high GOR oil reservoir which had ceased to flow due to high water cut or unavailability of enough reservoir support. Please see below production parameters of this well before and after the trial.

Table 3—Production parameters assessment

Parameters	Before (2004 data)	After (2018 data)
Gross	1050 blpd	1150 criteria
Net Oil	787 bopd	1030 bopd
BSW	25%	10%
Produced Gas	5 mmscf/d	4 mmscf/d
GOR	9500 scf/bbl	6000 scf/bbl
Lift Gas	1.5 mmscf/d	1.5 mmscf/d

The below plot was recorded on the real time pressure gauge during the unloading phase of this well.

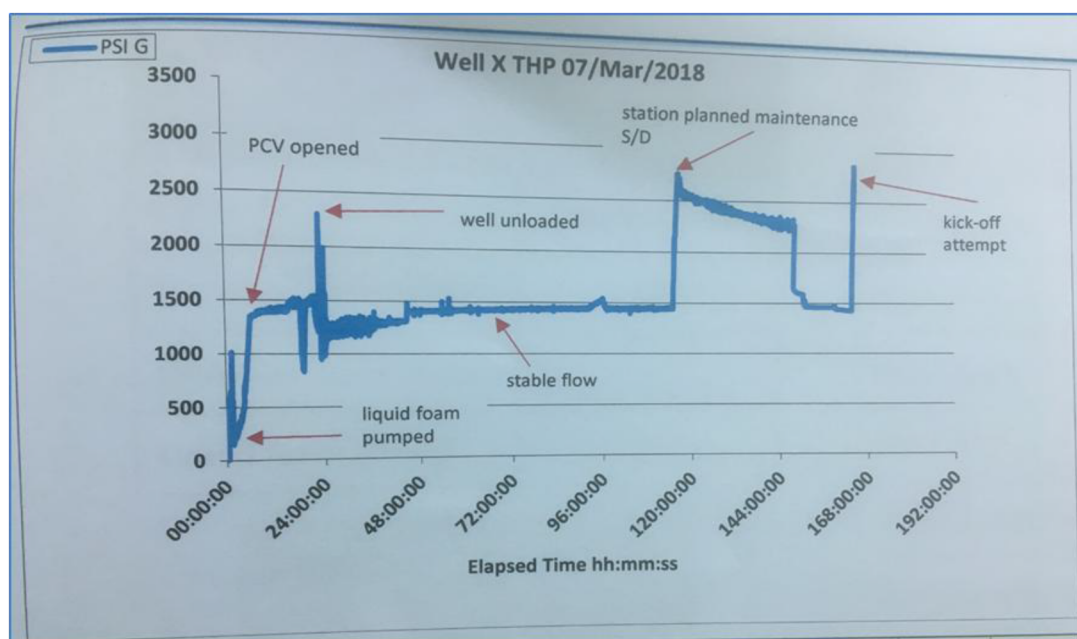


Figure 3—Well X unloading and production plot

The summary table below comprises results of five wells which were executed during this project.

Table 4—Well unloading and production summary

Well No.	Activated	No. of sticks dropped/ Vol. Liquid Foamer Inj.	Unlocked Gross Production	BS&W	Remarks
Well V	No	43/30 L	00 bpd	N/A	Operation suspended due to bad weather conditions
Well W	Yes	45/28 L	750 bpd	25%	Well produced for one month.
Well X	Yes	50/26 L	1150 bpd	10%	Well produced continuously post kick-off
Well Y	Yes	35/26 L	900 bpd	100%	Well produced water for 3 weeks. Well was shut-in.
Well Z	Yes	38/28 L	130 bpd	0%	Well activated and kept flowing for three hours. Shut-in due to the surface gas lift system shut down.

Conclusion

- Foam-assisted gas lift has proved to be an effective tool to revive high-salinity and high-GOR oil wells that had stopped production decades ago.
- Foam-assisted lift is a cost-effective technology which has played a significant role in unraveling the untapped reserves from existing wells.
- Enhancing well performance and recovering hydrocarbon resources are vital, often neglected tasks to improve field output, but this approach is limited by constraints from other wells and facilities. An integrated approach to closely monitoring the entire production system—wells, reservoirs and surface networks over time is the most efficient way to enhance overall field value.

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References

1. Gas well deliquification / James F. Lea, Henry V. Nickens, Mike R. Wells, Second Edition, 2008, pp. 4–6, Gulf Professional Publishing.
2. Analysis and Prediction of Minimum Flow Rate for the Continuous Removal of Liquids from Gas Wells/ Turner, R. G., Hubbard, M. G., & Dukler, A. E. (1969). *Journal of Petroleum Technology*.
3. A New Look at Predicting Gas-Well Load-Up / Coleman, S. B., Clay, H. B., McCurdy, D. G., & Norris, H. F. (1991). *SPE Production Engineering*.
4. Veeken, C., & Belfroid, S. (2010). New Perspective on Gas-Well Liquid Loading and Unloading. SPE Annual Technical Conference and Exhibition, Florence, Italy, September 19–22, 2010. SPE Paper No. SPE-134483-MS.
5. Willis, M.J., Horsup, D., & Nguyen, D. (2008). Chemical Foamers for Gas Well Deliquification. SPE Asia Pacific Oil and Gas Conference and Exhibition, Perth, Australia, October 20–22, 2008. SPE Paper No. SPE-115633-MS.